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Inventor(s)

SONG; Sungsub et al.

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### PUNCHING DEVICE AND PUNCHING METHOD USING SAME

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#### Abstract

A punching device including: a die to receive a substrate corresponding to a cutting target disposed thereon; a stripper spaced from and facing the die, and to move toward the substrate to pressurize and fix the substrate; and a punch to move toward the substrate to cut the substrate to form a scrap. The punch includes a first cut portion and a second cut portion having different shapes from each other with respect to a movement axis of the punch.

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**Inventors:** SONG; Sungsub (Yongin-si, KR), LEE; Jaewook (Yongin-si, KR), KIM; Du Hwan (Yongin-si, KR)

**Applicant:** SAMSUNG SDI CO., LTD. (Yongin-si, KR)

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#### Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Application No. 10-2024-0032261, filed on Mar. 6, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference herein.

## BACKGROUND

### 1. Field

[0002] Aspects of embodiments of the present disclosure relate to a punching device, and a punching method using the punching device.

### 2. Description of the Related Art

[0003] Unlike primary batteries that are not designed to be (re) charged, secondary (or rechargeable) batteries are batteries that are designed to be discharged and recharged. Low-capacity secondary batteries are used in portable, small electronic devices, such as smart phones, feature phones, notebook computers, digital cameras, and camcorders, while large-capacity secondary batteries are widely used as power sources for driving motors in hybrid vehicles and electric vehicles and for storing power (e.g., home and/or utility scale power storage). A secondary battery generally includes an electrode assembly composed of a positive electrode and a negative electrode, a case accommodating the same, and electrode terminals connected to the electrode assembly.

[0004] A positive electrode plate and a negative electrode plate of a lithium-ion battery is manufactured by applying a cathode active material to a positive electrode current collector in the form of a metal thin-film (e.g., an aluminum foil), and applying an anode active material to a negative electrode current collector in the form of a metal thin-film (e.g., a copper foil). The electrode plate is manufactured through mixing, coating, and pressing processes to manufacture a roll-shaped metal thin-film substrate coated with an active material, and slitting and notching processes to cut a roll-shaped metal thin-film substrate in accordance with desired battery specifications. The electrode plates for which the notching process has been completed are separated from each other by cutting between the electrode plates.

[0005] The above information disclosed in this Background section is for enhancement of understanding of the background of the present disclosure, and therefore, it may contain information that does not constitute related (or prior) art.

## SUMMARY

[0006] A punching device used in a notching process includes a punch including a push pin formed at an end portion thereof. A scrap formed when the electrode plate is cut may be pushed and discharged by the push pin. In this case, the push pin may come into contact with the electrode plate before the punch, causing damage to the electrode plate. In addition, a scrap that is not discharged may get caught in the punching device or may flow into a battery.

[0007] Embodiments of the present disclosure may be directed to a punching device, and a punching method using the punching device.

[0008] These and other aspects and features of the present disclosure will be described in or will be apparent from the following description of embodiments of the present disclosure.

[0009] According to one or more embodiments of the present disclosure, a punching device includes: a die configured to receive a substrate corresponding to a cutting target disposed thereon; a stripper spaced from and facing the die, and configured to move toward the substrate to pressurize and fix the substrate; and a punch configured to move toward the substrate to cut the substrate to form a scrap. The punch includes a first cut portion and a second cut portion having different shapes from each other with respect to a movement axis of the punch.

[0010] In an embodiment, the first cut portion and the second cut portion may be configured to cut a first portion and a second portion of the substrate, respectively, as the punch moves downward, and the scrap formed by cutting the first portion and the second portion may have a width between

the first portion and the second portion.

[0011] In an embodiment, a timing at which the first portion is cut and a timing at which the second portion is cut may be different from each other.

[0012] In an embodiment, the timing at which the first portion is cut and the timing at which the second portion is cut may be different from each other according to a height difference between sidewalls of the first cut portion and the second cut portion.

[0013] In an embodiment, a width between the first portion and the second portion may be four times or more the height difference between the sidewalls of the first cut portion and the second cut portion.

[0014] In an embodiment, the height difference between the sidewalls of the first cut portion and the second cut portion may be 0.8 mm to 1.2 mm.

[0015] In an embodiment, the width between the first portion and the second portion may be 2 mm to 6 mm.

[0016] In an embodiment, the first cut portion may include: a first sidewall parallel to the movement axis of the punch; and a first cut surface extending from the first sidewall in a width direction of the punch. The second cut portion may include: a second sidewall parallel to the movement axis of the punch; and a second cut surface extending from the second sidewall in the width direction of the punch.

[0017] In an embodiment, each of an angle between the first sidewall and the first cut surface of the first cut portion and an angle between the second sidewall and the second cut surface of the second cut portion may be 90 degrees.

[0018] In an embodiment, each of an angle between the first sidewall and the first cut surface of the first cut portion and an angle between the second sidewall and the second cut surface of the second cut portion may be less than 90 degrees, and the first cut surface and the second cut surface may be recessed toward an inside of the punch.

[0019] In an embodiment, each of an angle between the first sidewall and the first cut surface of the first cut portion and an angle between the second sidewall and the second cut surface of the second cut portion may be less than 90 degrees, and at least one of the first cut surface or the second cut surface may be recessed toward an inside of the punch.

[0020] In an embodiment, an angle between the first sidewall and the first cut surface of the first cut portion may be 90 degrees or more, and an angle between the second sidewall and the second cut surface of the second cut portion may be less than 90 degrees.

[0021] In an embodiment, the stripper may have a punch accommodation portion configured to accommodate the punch, and the punch accommodation portion may have a shape of a through-hole through which the punch may be configured to be inserted via the stripper.

[0022] In an embodiment, the die may include a discharge portion extending in a direction corresponding to the punch accommodation portion to communicate with the punch accommodation portion, and the scrap may be cut to be discharged through the discharge portion in an inclined state.

[0023] According to one or more embodiments of the present disclosure, a punching method includes: disposing a substrate corresponding to a cutting target on a die; lowering a stripper to pressurize and fix the substrate disposed on the die; cutting the substrate by using a punch; and discharging a scrap formed by the cutting of the substrate. The punch includes a first cut portion and a second cut portion having different shapes from each other with respect to a movement axis of the punch.

[0024] In an embodiment, a first height of a sidewall of the first cut portion may be different from a second height of a sidewall of the second cut portion.

[0025] In an embodiment, the cutting of the substrate may include: lowering the punch; cutting a first portion of the substrate by the first cut portion; and cutting a second portion of the substrate by the second cut portion. The first portion and the second portion may be cut at different timings

from each other according to a height difference between the first cut portion and the second cut portion, such that the scrap may be discharged in an inclined state.

[0026] In an embodiment, a width between the first portion and the second portion may be four times or more the height difference between the first cut portion and the second cut portion.

[0027] In an embodiment, the height difference between the first cut portion and the second cut portion may be 0.8 mm to 1.2 mm.

[0028] In an embodiment, a width between the first portion and the second portion may be 2 mm to 6 mm.

[0029] According to some embodiments of the present disclosure, a timing at which the substrate is cut by the first cut portion and a timing at which the substrate is cut by the second cut portion may be different from each other, such that the scrap may fall off in an inclined state. Accordingly, even in a case where the width of the discharge portion and the width of the scrap correspond to the same or approximate value, the scrap may be smoothly discharged through the discharge portion.

[0030] According to some embodiments of the present disclosure, by forming the angle of the contact region between the cut portion of the punch and the substrate at an acute angle, an area where the cut portion and the substrate are in contact with each other may be minimized or reduced. Accordingly, a structure for transmitting a shear force to the substrate may be formed.

[0031] According to some embodiments of the present disclosure, each of the angle formed by the first sidewall and the first cut surface and the angle formed by the second sidewall and the second cut surface may be formed at 90 degrees, such that the structure for pushing out the scrap may be formed.

[0032] According to some embodiments of the present disclosure, by appropriately changing the curvature of the first cut surface and/or the second cut surface that is recessed toward the inside of the punch, the structure for the first cut portion and/or the second cut portion to transmit a shear force to the substrate may be formed.

[0033] According to some embodiments of the present disclosure, problems that may occur in a case where the scrap is not smoothly discharged and is piled up in the punching device may be prevented. For example, low voltage defects, short circuit defects, and capacity defects, which may occur in a case where the scrap flows into the battery, may be prevented.

[0034] According to some embodiments of the present disclosure, the scrap may be smoothly discharged by simply changing the structure of the cut surface without expanding the diameter of the discharge portion.

[0035] However, aspects and features of the present disclosure are not limited to those described above, and other aspects and features not mentioned will be clearly understood by a person skilled in the art from the detailed description, described below.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0036] The following drawings attached to this specification illustrate embodiments of the present disclosure, and further describe aspects and features of the present disclosure together with the detailed description of the present disclosure. Thus, the present disclosure should not be construed as being limited to the drawings:

[0037] FIG. 1 illustrates an exploded perspective view of a punching device according to an embodiment of the present disclosure;

[0038] FIG. 2 illustrates an example of a punching device according to an embodiment of the present disclosure;

[0039] FIG. 3 illustrates an example of a punching device according to an embodiment of the present disclosure;

[0040] FIG. 4 illustrates an enlarged view showing a punch of FIG. 3 according to an embodiment of the present disclosure;

[0041] FIG. 5 illustrates an example of a punching device according to an embodiment of the present disclosure;

[0042] FIG. 6 illustrates an enlarged view showing a punch of FIG. 5 according to an embodiment of the present disclosure;

[0043] FIG. 7 illustrates an example of a punching device according to an embodiment of the present disclosure;

[0044] FIG. 8 illustrates an enlarged view showing a punch of FIG. 7 according to an embodiment of the present disclosure;

[0045] FIG. 9 illustrates an example of a punching device according to an embodiment of the present disclosure;

[0046] FIG. 10 illustrates an enlarged view showing a punch of FIG. 9 according to an embodiment of the present disclosure;

[0047] FIG. 11 illustrates a ratio between a height difference of a first cut portion and a second cut portion and a width of a scrap according to an embodiment of the present disclosure; and

[0048] FIG. 12 illustrates a flowchart showing an example of a punching method according to an embodiment of the present disclosure.

#### DETAILED DESCRIPTION

[0049] Hereinafter, embodiments of the present disclosure will be described, in detail, with reference to the accompanying drawings. The terms or words used in this specification and claims should not be construed as being limited to the usual or dictionary meaning and should be interpreted as meaning and concept consistent with the technical idea of the present disclosure based on the principle that the inventor can be his/her own lexicographer to appropriately define the concept of the term to explain his/her invention in the best way.

[0050] The embodiments described in this specification and the configurations shown in the drawings are only some of the embodiments of the present disclosure and do not represent all of the technical ideas, aspects, and features of the present disclosure. Accordingly, it should be understood that there may be various equivalents and modifications that can replace or modify the embodiments described herein at the time of filing this application.

[0051] It will be understood that when an element or layer is referred to as being “on,” “connected to,” or “coupled to” another element or layer, it may be directly on, connected, or coupled to the other element or layer or one or more intervening elements or layers may also be present. When an element or layer is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. For example, when a first element is described as being “coupled” or “connected” to a second element, the first element may be directly coupled or connected to the second element or the first element may be indirectly coupled or connected to the second element via one or more intervening elements.

[0052] In the figures, dimensions of the various elements, layers, etc. may be exaggerated for clarity of illustration. The same reference numerals designate the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Further, the use of “may” when describing embodiments of the present disclosure relates to “one or more embodiments of the present disclosure.” Expressions, such as “at least one of” and “any one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. When phrases such as “at least one of A, B and C,” “at least one of A, B or C,” “at least one selected from a group of A, B and C,” or “at least one selected from among A, B and C” are used to designate a list of elements A, B and C, the phrase may refer to any and all suitable combinations or a subset of A, B and C, such as A, B, C, A and B, A and C, B and C, or A and B and C. As used herein, the terms “use,” “using,” and “used” may be considered synonymous with the terms “utilize,” “utilizing,” and “utilized,” respectively. As used herein, the terms

“substantially,” “about,” and similar terms are used as terms of approximation and not as terms of degree, and are intended to account for the inherent variations in measured or calculated values that would be recognized by those of ordinary skill in the art.

[0053] It will be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers, and/or sections, these elements, components, regions, layers, and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer, or section from another element, component, region, layer, or section. Thus, a first element, component, region, layer, or section discussed below could be termed a second element, component, region, layer, or section without departing from the teachings of example embodiments.

[0054] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” or “over” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein should be interpreted accordingly.

[0055] The terminology used herein is for the purpose of describing embodiments of the present disclosure and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0056] Also, any numerical range disclosed and/or recited herein is intended to include all sub-ranges of the same numerical precision subsumed within the recited range. For example, a range of “1.0 to 10.0” is intended to include all subranges between (and including) the recited minimum value of 1.0 and the recited maximum value of 10.0, that is, having a minimum value equal to or greater than 1.0 and a maximum value equal to or less than 10.0, such as, for example, 2.4 to 7.6. Any maximum numerical limitation recited herein is intended to include all lower numerical limitations subsumed therein, and any minimum numerical limitation recited in this specification is intended to include all higher numerical limitations subsumed therein. Accordingly, Applicant reserves the right to amend this specification, including the claims, to expressly recite any sub-range subsumed within the ranges expressly recited herein. All such ranges are intended to be inherently described in this specification such that amending to expressly recite any such subranges would comply with the requirements of 35 U.S.C. § 112 (a) and 35 U.S.C. § 132 (a).

[0057] References to two compared elements, features, etc. as being “the same” may mean that they are “substantially the same”. Thus, the phrase “substantially the same” may include a case having a deviation that is considered low in the art, for example, a deviation of 5% or less. In addition, when a certain parameter is referred to as being uniform in a given region, it may mean that it is uniform in terms of an average.

[0058] Throughout the specification, unless otherwise stated, each element may be singular or plural.

[0059] Arranging an arbitrary element “above (or below)” or “on (under)” another element may mean that the arbitrary element may be disposed in contact with the upper (or lower) surface of the element, and another element may also be interposed between the element and the arbitrary

element disposed on (or under) the element.

[0060] In addition, it will be understood that when a component is referred to as being “linked,” “coupled,” or “connected” to another component, the elements may be directly “coupled,” “linked” or “connected” to each other, or another component may be “interposed” between the components”.

[0061] Throughout the specification, when “A and/or B” is stated, it means A, B or A and B, unless otherwise stated. That is, “and/or” includes any or all combinations of a plurality of items enumerated. When “C to D” is stated, it means C or more and D or less, unless otherwise specified.

[0062] FIG. 1 illustrates an exploded perspective view of a punching device **100** according to an embodiment of the present disclosure.

[0063] The punching device **100** according to an embodiment of the present disclosure may be a punching mold that may blank a continuously provided metal thin-film substrate, and cut the thin-film substrate into electrode plates satisfying a desired standard (e.g., a predetermined standard). In some embodiments, the punching device **100** may be a punching mold that may cut between the electrode plates in order to separate the electrode plates that are cut in accordance with standard (e.g., the predetermined standard).

[0064] Referring to FIG. 1, the punching device **100** may include a first mold **110** and a second mold **120**. The first mold **110** may refer to an upper mold. The second mold **120** may refer to a lower mold. Hereinafter, an upper portion may be a direction toward the first mold **110** (or a portion thereof), and a lower portion may be a direction toward the second mold **120** (or a portion thereof). In an embodiment, a substrate coated with an active material may be provided to the punching device **100**.

[0065] The first mold **110** may include a first base plate **112**, a stripper plate **114**, a punch **116**, and a stripper **118**. The stripper plate **114** may be fixedly connected to (e.g., coupled to or attached to) the lower portion of the first base plate **112**. The stripper **118** may be connected to (e.g., coupled to or attached to) the stripper plate **114**. For example, a stripper connection pin may connect the stripper **118** to the stripper plate **114**. The stripper **118** may include a punch accommodation portion **118a**. The punch **116** may be connected to (e.g., coupled to or attached to) the stripper plate **114**, and may be accommodated in the punch accommodation portion **118a** of the stripper **118**.

[0066] A cut portion of an asymmetric shape may be formed on the lower surface of the punch **116** with respect to a direction of a movement axis of the punch **116**. Some examples of the shape of the punch **116** are described in more detail below with reference to FIGS. 3 to 10.

[0067] The second mold **120** may include a second base plate **124** and a die **122**. The die **122** may be fixedly connected to (e.g., coupled to or attached to) the second base plate **124**. The die **122** may be fixedly connected to (e.g., coupled to or attached to) the upper portion of the second base plate **124**. The die **122** may include a discharge portion **122a**.

[0068] The first mold **110** and the second mold **120** may be spaced apart from each other, and may be connected to each other. For example, a base guide post may connect the first base plate **112** to the second base plate **124**, such that the first mold **110** and the second mold **120** are connected to (e.g., coupled to or attached to) each other. However, the present disclosure is not limited thereto, and the punching device may additionally include other components in the coupling structure between the first mold **110** and the second mold **120** for cutting the metal thin-film substrate.

[0069] The first mold **110** may descend or ascend toward the second mold **120**. In a case where the first mold **110** descends, the stripper **118** may come into contact with the die **122**. In a case where the metal thin-film substrate (e.g., the electrode plate) is disposed on the die **122**, the stripper **118** may be fixed to be in contact with the substrate (e.g., the electrode plate). In some embodiments, in a case where the first mold **110** descends, the punch **116** may descend and be inserted into the discharge portion **122a**. In a case where the metal thin-film substrate (e.g., the electrode plate) is disposed on the die **122**, the punch **116** may cut the metal thin-film substrate (e.g., the electrode plate). A scrap formed in a case where the metal thin-film substrate (e.g., the electrode plate) is cut

may be discharged through the discharge portion **122a**.

[0070] The discharge portion **112a** and/or the punch **116** may be formed in a shape corresponding to a desired standard (e.g., a predetermined standard). For example, the discharge portion **112a** and/or the punch **116** may correspond to the cut shape of the metal thin-film substrate. Similarly, the punch accommodation portion **118a** may be formed to correspond to the shape of the punch to accommodate the punch **116**.

[0071] Referring to FIG. **1**, one side of the punch accommodation portion **118a** is shown as an open structure, but the present disclosure is not limited thereto. For example, the punch accommodation portion **118a** may be in the form of a through-hole through which the punch **116** is inserted via the stripper **118**.

[0072] In addition to the above, the punching device **100** illustrated in FIG. **1** may further include a configuration used for manufacturing an electrode plate for a secondary battery.

[0073] FIG. **2** illustrates an example of a punching device **200** according to an embodiment of the present disclosure.

[0074] In an embodiment, the punching device **200** may include a die **210**, a stripper **220**, and a punch **230**. The stripper **220** may be spaced apart from the upper portion of the die **210** so as to face the die **210**. The stripper **220** may include a punch accommodation portion **222** for accommodating the punch **230**. The punch **230** may move upward or downward through the punch accommodation portion **222**.

[0075] In an embodiment, punch **230** may include a push pin **232**. The push pin **232** may be formed on the lower surface of the punch **230**, and may first contact the substrate **240** disposed on the die **210** in a case where the punch **230** moves downward. The substrate **240** may refer to a metal thin-film before being coated with an active material. The substrate **240** may refer to an electrode plate coated with an active material.

[0076] In an embodiment, the punching device **200** may cut the substrate **240**. The substrate **240** corresponding to a cutting target may be disposed on the upper surface of the die **210**. Thereafter, the stripper **220** may move downward toward the substrate **240**, and pressurize and fix the substrate **240**. FIG. **2** illustrates an example in which the substrate **240** is pressurized and fixed in a state where the stripper **220** moves downward.

[0077] Thereafter, the punch **230** may move downward through the punch accommodation portion **222**. In this case, the push pin **232** formed on the lower surface of the punch **230** may first contact the substrate **240**. In a case where the punch **230** additionally moves downward, the push pin **232** may be drawn into the inside of the punch **230**, and the punch **230** and the substrate **240** come into contact with each other, causing the substrate **240** to be cut. A scrap formed in a case where the substrate **240** is cut may be pushed by the push pin **232**, and discharged through the discharge portion **212**.

[0078] In a case where the scrap formed when the substrate is cut is not discharged properly, the scrap may get caught in the punching device, or piles up in the discharge portion may be caused. In this case, the scrap may flow into the battery, causing low voltage defects, short circuit defects, and capacity defects. As such, the push pin **232** may be formed on the lower surface of the punch **230**, as shown in the example of FIG. **2**. However, even in this case, the scrap may fall off vertically through the discharge portion **212**. The push pin **232** may first contact the substrate **240** before the punch **230** contacts the substrate **240**, causing a risk of damage (or damage) to the substrate **240**.

[0079] FIG. **3** illustrates an example of a punching device **300** according to an embodiment of the present disclosure. FIG. **4** illustrates an enlarged view showing a punch **330** of FIG. **3** according to an embodiment of the present disclosure.

[0080] Referring to FIG. **3**, the punching device **300** may include a die **310**, a stripper **320**, and a punch **330**. The stripper **320** may be spaced apart from the upper portion of the die **310** so as to face the die **310**. The stripper **320** may include a punch accommodation portion **322** for accommodating the punch **330**. The punch **330** may move upward or downward through the punch



accommodation portion **322**.

[0081] In an embodiment, the punching device **300** may cut the substrate **340**. In some embodiments, a substrate **340** corresponding to a cutting target may be disposed on the upper surface of the die **310**. Thereafter, the stripper **320** may move downward toward the substrate **340**, and may pressurize and fix the substrate **340**. In some embodiments, the punch **330** may cut the substrate **340** by moving downward, so as to pass through the punch accommodation portion **322**. A scrap formed in a case where the substrate **340** is cut may be discharged through a discharge portion **312** formed in the die **310**.

[0082] In an embodiment, the discharge portion **312** and/or the punch **330** may be formed in a suitable shape corresponding to a desired standard (e.g., a predetermined standard). For example, the discharge portion **312** and/or the punch **330** may have a shape corresponding to the cut shape of the substrate **340**. Similarly, the punch accommodation portion **322** may be formed to correspond to the shape of the punch **330** so as to accommodate the punch **330**. Accordingly, the shape of the scrap formed by the punch **330** may also correspond to the shape of the discharge portion **312**.

[0083] In an embodiment, the punch **330** may include cut portions that have asymmetrical shapes with respect to the direction of the movement axis **350** of the punch **330**. For example, a first cut portion of the punch **330** may correspond to the left side of the punch **330** with respect to the direction of the movement axis **350** of the punch **330**, and a second cut portion of the punch **330** may correspond to the right side with respect to the direction of the movement axis **350** of the punch **330**. Both sidewalls of the first cut portion and the second cut portion may be formed to have different heights from each other. Accordingly, the timings at which the substrate **340** is cut by the cut portions on opposite sides of the punch **330** may be different from each other. For example, the timing at which a first portion **342** of the substrate **340** is cut by the first cut portion of the punch **330** may be different from the timing at which a second portion **344** of the substrate **340** is cut by the second cut portion of the punch **330**.

[0084] FIG. 4 illustrates an enlarged view of the punch **330** of FIG. 3. Referring to FIG. 4, the punch **330** may include the first cut portion **410** and the second cut portion **420** formed in different shapes from each other. The first cut portion **410** and the second cut portion **420** may be arranged side-by-side with each other in the width direction of the punch **330**.

[0085] In an embodiment, the first cut portion **410** may include a first sidewall **412** parallel to or substantially parallel to the movement axis **350** of the punch **330**, and a first cut surface **414** extending from the first sidewall **412**. The first cut surface **414** may be bent and extended from the first sidewall **412** in the width direction of the punch **330**, while forming an angle (e.g., a certain or predetermined angle) therebetween. The angle formed between the first sidewall **412** and the first cut surface **414** may be less than 90 degrees.

[0086] Similarly, the second cut portion **420** may include a second sidewall **422** parallel to or substantially parallel to the movement axis **350** of the punch **330**, and a second cut surface **424** bent and extended from the second sidewall **422** in the width direction of the punch **330** while forming an angle (e.g., a certain or predetermined angle) therebetween. The angle formed between the second sidewall **422** and the second cut surface **424** may be less than 90 degrees. The first cut surface **414** and the second cut surface **424** may each constitute a plane, and may each form a shape or a surface that is recessed toward the inside of the punch **330** with respect to the movement axis **350** of the punch **330**.

[0087] In an embodiment, the height of the first sidewall **412** of the first cut portion **410** may be different from the height of the second sidewall **422** of the second cut portion **420**. Accordingly, a height difference ( $\Delta h$ ) may be formed between the first sidewall **412** and the second sidewall **422**.

[0088] In an embodiment, the timing at which the substrate is cut by the first cut portion **410** may be different from the timing at which the substrate is cut by the second cut portion **420**. In some embodiments, as the punch **330** moves downward, the second cut portion **420**, which has a higher sidewall height, may first cut the substrate while coming into contact with the substrate. Thereafter,

as the punch **330** additionally moves downward, the first cut portion **410** may cut the substrate while coming into contact with the substrate. In some embodiments, after the second portion of the substrate (e.g., the second portion **344** in FIG. **3**) is cut by the second cut portion **420**, the first portion of the substrate (e.g., the first portion **342** in FIG. **3**) may be cut by the first cut portion **410**. In this case, a scrap having a width (w) between the first portion and the second portion may be formed.

[0089] FIG. **4** illustrates that the height of the second cut portion **420** is greater (e.g., higher or larger) than the height of the first cut portion **410**, but the present disclosure is not limited thereto, and the height of the first cut portion **410** may be greater (e.g., higher or larger) than the height of the second cut portion **420**. In some embodiments, after the first portion of the substrate (e.g., the first portion **342** in FIG. **3**) is cut by the first cut portion **410**, the second portion of the substrate (e.g., the second portion **344** of FIG. **3**) may be cut by the second cut portion **420**.

[0090] FIG. **4** illustrates that a width ratio of the first cut portion **410** to the second cut portion **420** is 1:1, but the present disclosure is not limited thereto. For example, in some embodiments, the height difference ( $\Delta h$ ) may be formed between the first cut portion **410** and the second cut portion **420**, each of the angle formed by the first sidewall **412** and the first cut surface **414** and the angle formed by the second sidewall **422** and the second cut surface **424** may be less than 90 degrees, and the widths of the two cut portions may be appropriately modified as needed or desired.

[0091] FIG. **4** illustrates that the first cut surface **414** and the second cut surface **424** are connected to each other in a V-shape, but the present disclosure is not limited thereto. For example, the inclination of the first cut surface **414** and/or the second cut surface **424** may be variously modified at a suitable point (e.g., at a specific or predetermined point) inside the punch **330**. In some embodiments, each of the angle formed by the first sidewall **412** and the first cut surface **414** and the angle formed by the second sidewall **422** and the second cut surface **424** forms an acute angle, and the structure of the cut surface may be variously modified.

[0092] As such, the timing at which the substrate is cut by the first cut portion **410** and the timing at which the substrate is cut by the second cut portion **420** may be different from each other, such that the scrap may fall off in an inclined state. Accordingly, even in a case where the width of the discharge portion and the width of the scrap correspond to the same or approximate value as each other, the scrap may be smoothly discharged through the discharge portion.

[0093] In some embodiments, by forming the angle of the contact region between the cut portion of the punch and the substrate at an acute angle, the area where the cut portion and the substrate are in contact with each other may be minimized or reduced. Accordingly, a structure for transmitting a shear force to the substrate may be formed.

[0094] FIG. **5** illustrates an example of a punching device **500** according to an embodiment of the present disclosure. FIG. **6** illustrates an enlarged view showing a punch **530** of FIG. **5** according to an embodiment of the present disclosure.

[0095] Referring to FIG. **5**, the punching device **500** may include a die **510**, a stripper **520**, and a punch **530**. Hereinafter with reference to FIGS. **5** and **6**, redundant description as those described above with respect to the other figures may not be repeated.

[0096] Referring to FIGS. **5** and **6**, the punch **530** may include a first cut portion **610** and a second cut portion **620** formed in different shapes from each other. The first cut portion **610** and the second cut portion **620** may be arranged side-by-side with each other in the width direction of the punch **530**.

[0097] In an embodiment, the first cut portion **610** may include a first sidewall **612** parallel to or substantially parallel to the movement axis **550** of the punch **530**, and a first cut surface **614** extending from the first sidewall **612**. The first cut surface **614** may extend from the first sidewall **612** in the width direction of the punch **530**. The angle formed between the first sidewall **612** and the first cut surface **614** may be 90 degrees.

[0098] Similarly, the second cut portion **620** may include a second sidewall **622** parallel to or

substantially parallel to the movement axis **550** of the punch **530**, and a second cut surface **624** extending from the second sidewall **622** in the width direction of the punch **530**. The angle formed between the second sidewall **622** and the second cut surface **624** may be 90 degrees.

[0099] In an embodiment, the height of the first sidewall **612** of the first cut portion **610** may be different from the height of the second sidewall **622** of the second cut portion **620**. Accordingly, a height difference ( $\Delta h$ ) may be formed between the first sidewall **612** and the second sidewall **622**.

[0100] In an embodiment, a step may be formed between the first cut surface **614** and the second cut surface **624**. For example, by forming the height difference ( $\Delta h$ ) between the first sidewall **612** and the second sidewall **622**, a step having the height difference ( $\Delta h$ ) may be formed between the first cut surface **614** and the second cut surface **624**.

[0101] The first cut surface **614** and the first sidewall **612** form an angle of 90 degrees up to a point (e.g., a specific or predetermined point) inside the punch **530**, and the inclination of the surface after the point may be changed. Similarly, the first cut surface **624** and the first sidewall **622** form an angle of 90 degrees up to a point (e.g., a specific or predetermined point) inside the punch **530**, and the inclination of the surface after the point may be changed. In some embodiments, each of the angle formed by the first sidewall **612** and the first cut surface **614** and the angle formed by the second sidewall **622** and the second cut surface **624** forms 90 degrees, and the structure of the cut surface may be variously modified as needed or desired.

[0102] In an embodiment, the timing at which the substrate is cut by the first cut portion **610** may be different from the timing at which the substrate is cut by the second cut portion **620**. In some embodiments, as the punch **530** moves downward, the second cut portion **620**, which has a greater (e.g., a higher or a larger) sidewall height, may cut the substrate while coming into contact with the substrate. Thereafter, as the punch **530** additionally moves downward, the first cut portion **610** may cut the substrate while coming into contact with the substrate. In some embodiments, after the second portion of the substrate (e.g., the second portion **544** in FIG. 5) is cut by the second cut portion **620**, the first portion of the substrate (e.g., the first portion **542** of FIG. 5) may be cut by the first cut portion **610**. In this case, a scrap having a width ( $w$ ) between the first portion and the second portion may be formed.

[0103] Similarly to that described above with reference to FIG. 4, FIG. 6 illustrates that the height of the second cut portion **620** is greater (e.g., higher or larger) than the height of the first cut portion **610**, but the present disclosure is not limited thereto, and the height of the first cut portion **610** may be greater (e.g., higher or larger) than the height of the second cut portion **620**. FIG. 6 also illustrates that the width ratio of the first cut portion **610** to the second cut portion **620** is 1:1, but the present disclosure is not limited thereto. For example, the height difference ( $\Delta h$ ) may be formed between the first cut portion **610** and the second cut portion **620**, each of the angle formed by the first sidewall **612** and the first cut surface **614** and the angle formed by the second sidewall **622** and the second cut surface **624** may 90 degrees or less, and the widths of the two cut portions may be appropriately modified as needed or desired.

[0104] In some embodiments, each of the angle formed by the first sidewall **612** and the first cut surface **614** and the angle formed by the second sidewall **622** and the second cut surface **624** forms 90 degrees, and the structure of the cut surface may be variously modified as needed or desired.

[0105] FIG. 7 illustrates an example of a punching device **700** according to an embodiment of the present disclosure. FIG. 8 illustrates an enlarged view of a punch **730** of FIG. 7 according to an embodiment of the present disclosure.

[0106] Referring to FIG. 7, the punching device **700** may include a die **710**, a stripper **720**, and a punch **730**. Hereinafter with reference to FIGS. 7 and 8, redundant description as those described above with respect to the other figures may not be repeated.

[0107] Referring to FIGS. 7 and 8, the punch **730** may include a first cut portion **810** and a second cut portion **820** formed in different shapes from each other. The first cut portion **810** and the second cut portion **820** may be arranged side-by-side with each other in the width direction of the punch

**730.**

[0108] In an embodiment, the first cut portion **810** may include a first sidewall **812** parallel to or substantially parallel to a movement axis **750** of the punch **730**, and a first cut surface **814** extending from the first sidewall **812** in the width direction of the punch **730**. The first cut surface **814** may be in the form of a curved surface that is recessed toward the inside of the punch **730** with respect to the movement axis **750** of the punch **730**. Accordingly, the angle formed between the first sidewall **812** and the first cut surface **814** may be less than 90 degrees. In some embodiments, the slope of the tangent to the first cut surface **814** at the point where the first sidewall **812** and the first cut surface **814** are connected to each other may be less than 90 degrees.

[0109] Similarly, the second cut portion **820** may include a second sidewall **822** parallel to or substantially parallel to the movement axis **750** of the punch **730**, and a second cut surface **824** extending from the second sidewall **822** in the width direction of the punch **730**. The second cut surface **824** may be in the form of a curved surface that is recessed toward the inside of the punch **730** with respect to the movement axis **750** of the punch **730**. Accordingly, the angle formed between the second sidewall **822** and the second cut surface **824** may be less than 90 degrees. In some embodiments, the slope of the tangent of the second cut surface **824** at the point where the second sidewall **822** and the second cut surface **824** are connected to each other may be less than 90 degrees.

[0110] In an embodiment, the height of the first sidewall **812** of the first cut portion **810** may be different from the height of the second sidewall **822** of the second cut portion **820**. Accordingly, a height difference ( $\Delta h$ ) may be formed between the first sidewall **812** and the second sidewall **822**.

[0111] In an embodiment, the timing at which the substrate is cut by the first cut portion **810** may be different from the timing at which the substrate is cut by the second cut portion **820**. In some embodiments, as the punch **730** moves downward, the second cut portion **820**, which has a greater (e.g., a higher or a larger) sidewall height, may cut the substrate while coming into contact with the substrate. Thereafter, as the punch **730** additionally moves downward, the first cut portion **810** may cut the substrate while coming into contact with the substrate. In some embodiments, after the second portion of the substrate (e.g., the second portion **744** in FIG. 7) is cut by the second cut portion **820**, the first portion of the substrate (e.g., the first portion **742** of FIG. 7) may be cut by the first cut portion **810**. In this case, a scrap having a width ( $w$ ) between the first portion and the second portion may be formed.

[0112] FIG. 8 illustrates that the first cut surface **814** and the second cut surface **824** are smoothly connected to each other without a step therebetween, but the present disclosure is not limited thereto. In some embodiments, a step may be formed between the first cut surface **814** and the second cut surface **824**. In some embodiments, the first cut surface **814** and the second cut surface **824** may be formed in the form of a curved surface up to a point (e.g., a specific or predetermined point) inside the punch **730**, and the surface after the point may be connected in the form of a flat or substantially flat surface. Each of the first cut surface **814** and the second cut surface **824** forms a curved surface recessed toward the inside of the punch **730** in the vicinity extending from the first sidewall **812** and the second sidewall **822**, and the structure of the cut surface may be variously modified as needed or desired.

[0113] In an embodiment, the first cut surface **814** may form a curved surface in the vicinity extending from the first sidewall **812**, and the second cut surface **824** may form a flat or substantially flat surface in the vicinity extending from the second sidewall **822**. In other embodiments, the second cut surface **824** may form a curved surface in the vicinity extending from the second sidewall **822**, and the first cut surface **814** may form a flat or substantially flat surface in the vicinity extending from the first sidewall **812**. In some embodiments, at least one of the first cut surface **814** or the second cut surface **824** may form a curved surface.

[0114] By appropriately modifying the curvature of the first cut surface **814** and/or the second cut surface **816** that is recessed toward the inside of the punch **730**, the first cut portion **810** and/or the

second cut portion **820** may form a structure for transmitting a shear force to the substrate.

[0115] FIG. **9** illustrates an example of a punching device **900** according to an embodiment of the present disclosure. FIG. **10** illustrates an enlarged view showing a punch **930** of FIG. **9** according to an embodiment of the present disclosure.

[0116] Referring to FIG. **9**, the punching device **900** may include a die **910**, a stripper **920**, and a punch **930**. Hereinafter with reference to FIGS. **9** and **10**, redundant description as those described above with respect to the other figures may not be repeated.

[0117] Referring to FIGS. **9** and **10**, the punch **930** may include a first cut portion **1010** and a second cut portion **1020** formed in different shapes from each other. The first cut portion **1010** and the second cut portion **1020** may be arranged side-by-side with each other in the width direction of the punch **930**.

[0118] In an embodiment, the first cut portion **1010** may include a first sidewall **1012** parallel to or substantially parallel to a movement axis **950** of the punch **930**, and a first cut surface **1014** extending from the first sidewall **1012** in the width direction of the punch **930**. The angle formed between the first sidewall **1012** and the first cut surface **1014** may be 90 degrees or more.

[0119] Similarly, the second cut portion **1020** may include a second sidewall **1022** parallel to or substantially parallel to the movement axis **950** of the punch **930**, and a second cut surface **1024** extending from the second sidewall **1022** in the width direction of the punch **930**. The angle formed between the second sidewall **1022** and the second cut surface **1024** may be less than 90 degrees.

[0120] In an embodiment, the height of the first sidewall **1012** of the first cut portion **1010** may be different from the height of the second sidewall **1022** of the second cut portion **1020**. Accordingly, a height difference ( $\Delta h$ ) may be formed between the first sidewall **1012** and the second sidewall **1022**.

[0121] In an embodiment, the timing at which the substrate is cut by the first cut portion **1010** may be different from the timing at which the substrate is cut by the second cut portion **1020**. In some embodiments, as the punch **930** moves downward, the second cut portion **1020**, which has a greater (e.g., a higher or a larger) sidewall height, may cut the substrate while coming into contact with the substrate. Thereafter, as the punch **930** additionally moves downward, the first cut portion **1010** may cut the substrate while coming into contact with the substrate. In some embodiments, after the second portion of the substrate (e.g., the second portion **944** in FIG. **9**) is cut by the second cut portion **1020**, the first portion of the substrate (e.g., the first portion **942** of FIG. **9**) may be cut by the first cut portion **1010**. In this case, a scrap having a width ( $w$ ) between the first portion and the second portion may be formed.

[0122] FIG. **10** illustrates that the angle formed by the first sidewall **1012** and the first cut surface **1014** is greater than 90 degrees, and the angle formed by the second sidewall **1022** and the second cutting surface **1024** is less than 90 degrees, but the present disclosure is not limited thereto. In some embodiments, one of the angle formed by the first sidewall **1012** and the first cut surface **1014** or the angle formed by the second sidewall **1022** and the second cut surface **1024** is less than 90 degrees, and the other thereof is 90 degrees or more. Therefore, the structure of the cut surface may be variously modified as needed or desired.

[0123] FIG. **11** illustrates a ratio between a height difference ( $\Delta h$ ) of a first cut portion and a second cut portion and a width ( $w$ ) of a scrap according to an embodiment of the present disclosure.

[0124] In an embodiment, a punching device **1100** may include a die having a discharge portion for discharging a scrap, a stripper having a punch accommodation portion, and a punch. In an embodiment, the punching device **1100** may cut a substrate. In some embodiments, the stripper pressurizes and fixes the substrate disposed on the upper surface of the die, and the punch moves downward so as to pass through the punch accommodation portion, thereby cutting the substrate. A scrap formed in a case where the substrate is cut may be discharged through the discharge portion.

[0125] In an embodiment, the discharge portion and/or the punch may be formed in a shape corresponding to a desired standard (e.g., a predetermined standard). For example, the discharge

portion and/or the punch may correspond to the cut shape of the substrate. Similarly, the punch accommodation portion may be formed to correspond to the shape of the punch so as to accommodate the punch. Accordingly, the shape of the scrap formed by the punch may also correspond to the shape of the discharge portion.

[0126] In an embodiment, the punch has two sidewalls of different heights with respect to a direction of a movement axis of the punch, and thus, the timings at which the substrate is cut by the cut portions on opposite sides of the punch may be different from each other. Accordingly, the scrap formed by cutting the substrate may fall off in an inclined state.

[0127] In an embodiment, the orthographic length (x) of the scrap width may be calculated according to Equation 1 below using the Pythagorean theorem. In Equation 1, x is the orthogonal length of the scrap width, Δh is the difference in height of the sidewalls of the cut portions on opposite sides, and w is the scrap width. The orthographic length (x) of the scrap width may refer to the orthographic length of the scrap with respect to a plane perpendicular to or substantially perpendicular to the movement axis of the punch.

$$[00001] \quad x^2 = w^2 - h^2 \quad \text{Equation 1}$$

[0128] Referring to Equation 1, in a case where the height difference (Δh) in the sidewalls between the cut portions on opposite sides is 1 mm, examples of the scrap width (w), the orthographic length (x) of the scrap width, and the difference (w-x) between the scrap width and the orthographic length of the scrap width are shown in Table 1 below. However, the present disclosure is not limited thereto, and the height difference (Δh) in the sidewalls may have any suitable value from 0.8 mm to 1.2 mm. In some embodiments, the height difference (Δh) in the sidewalls may have a value out of (e.g., different from) the corresponding range.

TABLE-US-00001

| TABLE 1 | Δh(mm) | w(mm)      | x(mm)      | w - x(mm) |
|---------|--------|------------|------------|-----------|
| 1       | 2      | about 1.73 | about 0.27 | 1         |
| 3       | 4      | about 2.83 | about 0.17 | 1         |
| 5       | 6      | about 3.87 | about 0.13 | 1         |
| 7       | 8      | about 4.90 | about 0.1  | 1         |
| 9       | 10     | about 5.92 | about 0.08 | 1         |
| 11      | 12     | about 6.93 | about 0.07 | 1         |

[0129] Referring to Table 1, the ratio (w/Δh) of the scrap width (w) to the height difference (Δh) in the sidewalls between the cut portions on opposite sides formed by cutting the substrate may be 2 to 7. For example, the ratio (w/Δh) of the scrap width (w) to the height difference (Δh) in the sidewalls between the cut portions on opposite sides may be 5. In this case, in a case assuming that the scrap width (w) is 5 mm and the height difference (Δh) in the sidewalls between the cut portion on opposite sides is 1 mm, the orthographic length (x) of the scrap width is about 4.90 mm, which is shorter than the scrap width (w), and the minor axis length (w-x) may be about 0.1 mm. In some embodiments, as the scrap falls off in an inclined state, the orthographic length (x) of the scrap width may be shortened compared to the scrap width (w).

[0130] In Table 1, only cases where the ratio (w/Δh) of the scrap width (w) to the height difference (Δh) in the sidewalls between the two cut portions are 2, 3, 4, 5, 6, and 7 are shown, but the ratio of the scrap width (w) to the height difference (Δh) in the sidewalls between the cut portions on opposite sides may be appropriately modified as needed or desired by various variables.

[0131] In an embodiment, the height difference (Δh) in the sidewalls between the cut portions on opposite sides may be 0.8 mm to 1.2 mm. In this case, the scrap width (w) may be 2 mm to 6 mm. In an embodiment, the height difference (Δh) in the sidewalls between the cut portions on opposite sides may be 1 mm, and the scrap width (w) may be 5 mm. In this case, more effective scrap discharge may be possible. In a case where the ratio (w/Δh) of the scrap width (w) to the height difference (Δh) in the sidewalls between the cut portions on opposite sides is 5, more effective scrap discharge may be possible.

[0132] Due to the height difference (Δh) in the sidewalls between the cut portions on opposite sides, the orthographic length (x) of the scrap width may be shorter than the scrap width (w). However, the present disclosure is not limited to the aforementioned values, and the height difference (Δh) in the sidewalls between the cut portions on opposite sides and the scrap width (w)

may be appropriately modified by various suitable variables, such as the spacing between the electrode plates applied to the substrate, the size of the punch, and/or the like.

[0133] According to some embodiments described above, the scrap may be smoothly discharged through the discharge portion as the scrap falls off in an inclined state. Accordingly, cases where the scrap is not smoothly discharged and is piled up in the punching device may be prevented or substantially prevented. For example, low voltage defects, short circuit defects, and capacity defects, which may occur in a case where the scrap flows into the battery, may be prevented or substantially prevented.

[0134] FIG. 12 illustrates a flowchart showing an example of a punching method according to an embodiment of the present disclosure.

[0135] In an embodiment, a punching method 1200 may be performed by a punching device (e.g., the punching device 100 of FIG. 1) that includes a die, a stripper, and a punch.

[0136] The punching method 1200 may start, and the substrate corresponding to a cutting target may be disposed on the die (S1210). Thereafter, the punching device may lower the stripper to pressurize and fix the substrate disposed on the die (S1220).

[0137] In an embodiment, the punching device may cut the substrate by using the punch (S1230). The punch may include a first cut portion and a second cut portion formed in different shapes from each other with respect to the movement axis of the punch. For example, the height (e.g., a first height) of the sidewall of the first cut portion may be different from the height (e.g., a second height) of the sidewall of the second cut portion.

[0138] In an embodiment, the punching device may lower the punch so as to cut the substrate by using the punch. Thereafter, the punching device may cut the first portion of the substrate with the first cut portion and then cut the second portion of the substrate with the second cut portion. In some embodiments, the punching device may cut the second portion of the substrate with the second cut portion and then cut the first portion of the substrate with the first cut portion.

[0139] In some embodiments, the width (w) between the first portion and the second portion may be twice or more the height difference ( $\Delta h$ ) in the sidewalls between the first cut portion and the second cut portion. In some embodiments, the height difference ( $\Delta h$ ) between the first cut portion and the second cut portion may be 0.8 mm to 1.2 mm. In some embodiments, the width (w) between the first portion and the second portion may be 2 mm to 6 mm. In some embodiments, the height difference ( $\Delta h$ ) between the first cut portion and the second cut portion may be 1 mm, and the width (w) between the first portion and the second portion may be 5 mm.

[0140] In an embodiment, the punching device may discharge the scrap formed in a case where the substrate is cut (S1240), and the method 1200 may end. In this case, the first portion and the second portion are cut at different timings from each other due to the height difference ( $\Delta h$ ) between the first cut portion and the second cut portion, such that the scrap may be discharged in an inclined state.

[0141] Although the present disclosure has been described above with respect to embodiments thereof, the present disclosure is not limited thereto. Various modifications and variations can be made thereto by those skilled in the art within the spirit of the present disclosure and the equivalent scope of the appended claims.

## DESCRIPTION OF SOME REFERENCE SYMBOLS

[0142] 100: punching device [0143] 110: first mold [0144] 112: first base plate [0145] 114: stripper plate [0146] 116: punch [0147] 118: stripper [0148] 118a: punch accommodation portion [0149] 120: second mold [0150] 122: die [0151] 122a: discharge portion [0152] 122: second base plate

## Claims

1. A punching device comprising: a die configured to receive a substrate corresponding to a cutting target disposed thereon; a stripper spaced from and facing the die, and configured to move toward

the substrate to pressurize and fix the substrate; and a punch configured to move toward the substrate to cut the substrate to form a scrap, wherein the punch comprises a first cut portion and a second cut portion having different shapes from each other with respect to a movement axis of the punch.

2. The punching device as claimed in claim 1, wherein the first cut portion and the second cut portion are configured to cut a first portion and a second portion of the substrate, respectively, as the punch moves downward, and wherein the scrap formed by cutting the first portion and the second portion has a width between the first portion and the second portion.
3. The punching device as claimed in claim 2, wherein a timing at which the first portion is cut and a timing at which the second portion is cut are different from each other.
4. The punching device as claimed in claim 3, wherein the timing at which the first portion is cut and the timing at which the second portion is cut are different from each other according to a height difference between sidewalls of the first cut portion and the second cut portion.
5. The punching device as claimed in claim 4, wherein the width between the first portion and the second portion is four times or more the height difference between the sidewalls of the first cut portion and the second cut portion.
6. The punching device as claimed in claim 4, wherein the height difference between the sidewalls of the first cut portion and the second cut portion is 0.8 mm to 1.2 mm.
7. The punching device as claimed in claim 4, wherein the width between the first portion and the second portion is 2 mm to 6 mm.
8. The punching device as claimed in claim 1, wherein the first cut portion comprises: a first sidewall parallel to the movement axis of the punch; and a first cut surface extending from the first sidewall in a width direction of the punch, and wherein the second cut portion comprises: a second sidewall parallel to the movement axis of the punch; and a second cut surface extending from the second sidewall in the width direction of the punch.
9. The punching device as claimed in claim 8, wherein each of an angle between the first sidewall and the first cut surface of the first cut portion and an angle between the second sidewall and the second cut surface of the second cut portion is 90 degrees.
10. The punching device as claimed in claim 8, wherein each of an angle between the first sidewall and the first cut surface of the first cut portion and an angle between the second sidewall and the second cut surface of the second cut portion is less than 90 degrees, and wherein the first cut surface and the second cut surface are recessed toward an inside of the punch.
11. The punching device as claimed in claim 8, wherein each of an angle between the first sidewall and the first cut surface of the first cut portion and an angle between the second sidewall and the second cut surface of the second cut portion is less than 90 degrees, and wherein at least one of the first cut surface or the second cut surface is recessed toward an inside of the punch.
12. The punching device as claimed in claim 8, wherein an angle between the first sidewall and the first cut surface of the first cut portion is 90 degrees or more, and wherein an angle between the second sidewall and the second cut surface of the second cut portion is less than 90 degrees.
13. The punching device as claimed in claim 1, wherein the stripper has a punch accommodation portion configured to accommodate the punch, and wherein the punch accommodation portion has a shape of a through-hole through which the punch is configured to be inserted via the stripper.
14. The punching device as claimed in claim 13, wherein the die comprises a discharge portion extending in a direction corresponding to the punch accommodation portion to communicate with the punch accommodation portion, and wherein the scrap is cut to be discharged through the discharge portion in an inclined state.
15. A punching method comprising: disposing a substrate corresponding to a cutting target on a die; lowering a stripper to pressurize and fix the substrate disposed on the die; cutting the substrate by using a punch; and discharging a scrap formed by the cutting of the substrate, wherein the punch comprises a first cut portion and a second cut portion having different shapes from each other with



respect to a movement axis of the punch.

**16.** The punching method as claimed in claim 15, wherein a first height of a sidewall of the first cut portion is different from a second height of a sidewall of the second cut portion.

**17.** The punching method as claimed in claim 16, wherein the cutting of the substrate comprises: lowering the punch; cutting a first portion of the substrate by the first cut portion; and cutting a second portion of the substrate by the second cut portion, and wherein the first portion and the second portion are cut at different timings from each other according to a height difference between the first cut portion and the second cut portion, such that the scrap is discharged in an inclined state.

**18.** The punching method as claimed in claim 17, wherein a width between the first portion and the second portion is four times or more the height difference between the first cut portion and the second cut portion.

**19.** The punching method as claimed in claim 17, wherein the height difference between the first cut portion and the second cut portion is 0.8 mm to 1.2 mm.

**20.** The punching method as claimed in claim 17, wherein a width between the first portion and the second portion is 2 mm to 6 mm.

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